Quantum noise around the amplifier similariton: phase-sensitive shape modes, energy–phase locking and the limits of matched-mode squeezing

Authors: to be set by the PI
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A pulse amplified in a fibre with normal dispersion is attracted to a parabolic similariton. We collect analytical results on the quantum noise of this process, each checked by an independent implementation written by a second model family. In the leading, dispersionless theory every Legendre shape mode $\ell \geq 2$ acts as a phase-sensitive amplifier: photon-normalized intensity- and phase-shape variances grow as $G^{1/3}$ and $G^{5/3}$ with the power gain G, against G for a linear amplifier, with prefactors that oscillate in $\ln G$. Two measurable projections therefore collapse onto the energy fluctuation: a five-component projection of the single-shot spectral covariance has eigenvalue ratio $\mu_2/\mu_1 = \Theta(G^{-2/3})$, and the phase quadrature of a matched local oscillator becomes energy-locked with deficit $1 - r^2 = \Theta(G^{-2/3})$, while its variance grows as $(32n_{sp}/9)G\Phi_{NL}^2 \propto G^{7/3}$. Amplifier noise does not forbid squeezing: at short length the matched mode is squeezed despite full spontaneous emission when $\gamma \int P^2 / \int P > n_{sp} g$. At large gain its minimum variance grows as $G^{1/3}$. We give the steps that carry each derivation, the regime as bounds on dimensionless groups, what failed in verification, and what to measure.

I. INTRODUCTION

A pulse amplified in a normally dispersive fibre approaches a parabolic similariton, whose width grows as $G^{1/3}$ and whose chirp is linear [1–3]; the parameter dynamics of this attraction are published [4, 5]. Its noise has been computed numerically for similariton lasers [6] and measured as classical intensity noise of the total power at one gain, without quantum, phase or shape-resolved noise [7, 8], or in spectral slices [9]. We collect analytical results on the quantum noise of single-pass similariton amplification: how the attraction reshapes vacuum and amplified spontaneous emission (ASE) in the low-order shape modes (Sec. III), what this does to two measurable projections, the single-shot spectrum and a matched local oscillator (LO) (Sec. IV), and when the matched mode can be squeezed despite phase-insensitive gain (Sec. V). For each statement we give the steps that carry it, its regime and how it was checked.

The statements are two results of the project repository at its "promising" tier, R-0011 and R-0014 (Table I). Each was proposed by one model family and verified with independent code written by another; R-0011 also passed a clean-context audit. Neither is formalized or peer reviewed. Two statements of the original claim failed and are reported as failures.

The ingredients are known: the hydrodynamic modes of a parabolic (Thomas–Fermi) condensate, with spectrum $\propto \ell(\ell+1)$ [10], and the separation of fluctuations on an expanding background [11]; the quantum limit of linear amplifiers [12] and the Langevin theory of fibre noise [13]; Kerr conversion of amplitude noise into phase noise [14] and its correction from measured intensity [15]; pulsed Kerr squeezing read with a pulse-matched LO [16, 17]. Linearizing the published width equation already gives the decay of the breathing ($\ell = 2$) mode at rate $g/3$, so that rate is not new. Bounded searches by agents of both families did not find the noise exponents, the spectral and phase–energy laws or the short-length squeezing criterion; a two-mode Kerr amplifier with a related noise advantage has been proposed outside fibres [18]. These searches do not establish novelty.

II. MODEL AND SCALES

We use the scalar nonlinear Schrödinger equation with flat, unsaturated power gain g and its Langevin noise,

$$\partial_z A = -\tfrac{i\beta_2}{2}\partial_T^2 A + i\gamma|A|^2 A + \tfrac{g}{2}A + F, \qquad (1)$$

with $\beta_2 > 0$, $|A|^2$ in photons per unit time, $\langle F^\dagger F\rangle = g n_{sp}\delta$ and $\langle FF^\dagger\rangle = g(n_{sp}-1)\delta$, $\delta = \delta(z-z')\delta(T-T')$; this reservoir keeps the commutator. The seed is a coherent state of N_0 photons, $G = e^{gz}$, $\tau = \ln G$ and $N_{\rm out} = GN_0$. The noise is linearized about the bright mean field.

With $L_D = T_0^2/\beta_2$ and $L_{NL} = 1/(\gamma P_0)$ the groups are $\varepsilon_0 = \sqrt{L_{NL}/L_D}$ and $\Gamma = g\sqrt{L_D L_{NL}}$; $\zeta = z/\sqrt{L_D L_{NL}}$, so $\tau = \Gamma\zeta$, and $x = T/T_0$. At leading order in ε_0 the intensity ρ and the velocity $v = -\varepsilon_0\partial_x\phi$ obey $\rho_\zeta + (\rho v)_x = \Gamma\rho$ and $v_\zeta + vv_x = -\rho_x$. The parabola $\bar\rho = \rho_{\rm pk}(1-\xi^2)$, $\bar v = a'x/a$, with comoving time $\xi = x/a$, is an exact solution if $a'' = 2e^{\Gamma\zeta}/a^2$ (unit peak and half-width at launch); its asymptote is $a = A_* e^{\tau/3}$ with $A_*^3 = 18/\Gamma^2$. The peak nonlinear phase $\Phi_{NL} = \int \gamma P_{\rm pk}\, dz$ is $\frac{3}{2\Gamma A_*\varepsilon_0}(G^{2/3}-1)$ on the asymptote, and the local semiclassical parameter $\varepsilon_{\rm eff} = \varepsilon_0/\sqrt{ae^{\Gamma\zeta}}$ falls as $G^{-2/3}$.

Fluctuations are written $\delta|A|^2 = \rho_{\rm pk} n(\xi)$ and $\delta\arg A = p(\xi)$, with Legendre components n_ℓ, p_ℓ. Quadratures have vacuum covariance I. The matched LO is the output mean field. It reads the number quadrature $X_A = \delta N/\sqrt{N}$ and the phase quadrature $P_{\rm LO} = 2\sqrt{N}\,\bar p$, where $\bar p = p_0 - p_2/5$ is the intensity-weighted mean phase; C_{AA}, C_{PP}, C_{AP} is their covariance $C_{\rm LO}$ and r their correlation. The linearization needs the phase noise to be small, $\Phi_{NL}^2/N_0 \ll 1$, which is not uniform in gain.

Numbers are cited by run: [P] `2026-10-18-pi-chatgpt-independent-verification` (GPT 6 Pro); [E] `2026-10-19-gpt-e-similariton` (gpt-e); [V] `2026-10-19-claude-interactive.13-similariton-lo-verification` (claude-interactive.13); [A] `2026-10-11-gpt-a-highlights-audit-c2` (gpt-a); [Q] `2026-10-19-claude-interactive-gpt6pro-quick-repro` (claude-interactive); [S] `2026-10-08-claude-a-amp-sim-checks` (claude-a). Appendix G lists each

number with its run and agent.

III. SHAPE MODES ARE AMPLIFIED PHASE-SENSITIVELY

Statement (R-0011a). For every fixed $\ell \geq 2$ and input vacuum plus ASE, in leading parabolic hydrodynamics on the asymptote,

$$N_{\rm out}\mathrm{Var}\, n_\ell = \Theta(G^{1/3}), \qquad \varepsilon_0^2 N_{\rm out}\mathrm{Var}\, p_\ell = \Theta(G^{5/3}), \quad (2)$$

against $\Theta(G)$ for both quadratures of a linear amplifier. Θ means bounded above and below by positive constants times the power; the prefactors oscillate in $\ln G$, so they need not converge.

Crux. Linearize about any parabolic background at fixed ξ. The gain term cancels against the growth of $\rho_{\rm pk}$, the phase is carried along the background flow, and the pressure term becomes Legendre's operator, $\partial_\xi[(1-\xi^2)\partial_\xi]P_\ell = -\ell(\ell+1)P_\ell$, as for the modes of a one-dimensional Thomas–Fermi condensate [10, 11]. Each component then evolves on its own:

$$\frac{dp_\ell}{d\zeta} = \frac{e^{\Gamma\zeta}}{a}\frac{n_\ell}{\varepsilon_0}, \qquad \frac{d}{d\zeta}\frac{n_\ell}{\varepsilon_0} = -\frac{\ell(\ell+1)}{a^2}\,p_\ell. \quad (3)$$

Eliminating p_ℓ on the asymptote gives, with $' = d/d\tau$,

$$n_\ell'' + \tfrac{2}{3}n_\ell' + \tfrac{\ell(\ell+1)}{18}n_\ell = 0, \quad (4)$$

with roots $\frac{1}{3}[-1 \pm \sqrt{1-\ell(\ell+1)/2}]$: the energy ($\ell = 0$) is neutral, $\ell = 1$ is a double root, and every $\ell \geq 2$ decays as $G^{-1/3}$ while oscillating at $\nu_\ell = \frac{1}{3}\sqrt{\ell(\ell+1)/2 - 1}\,\ln G$ ($\nu_2 = \sqrt{2}/3$). By Eq. (3), $p_\ell' \propto e^{2\tau/3}n_\ell/\varepsilon_0$, so the phase response grows as $G^{1/3}/\varepsilon_0$. In the coordinates $u_\ell = G^{1/3}n_\ell$, $w_\ell = \varepsilon_0 G^{-1/3}p_\ell$ the drift is a constant matrix B_ℓ with $B_\ell^2 = -\nu_\ell^2 I$, and the propagator is a bounded rotation. Noise enters n_ℓ with variance $\propto 1/N(\tau) \propto e^{-\tau}$, which is $e^{-\tau/3}$ in u_ℓ: vacuum and early ASE dominate, and the covariance of (u_ℓ, w_ℓ) tends to a bounded, positive, oscillating matrix. Undoing the scaling, $\mathrm{Var}\, n_\ell \sim G^{-2/3}/N_0$ and $\mathrm{Var}\, p_\ell \sim G^{2/3}/(\varepsilon_0^2 N_0)$; multiplying by $N_{\rm out} = GN_0$ gives Eq. (2). Relative to linear amplification, the attraction de-amplifies intensity-shape noise by $G^{-2/3}$ and amplifies phase-shape noise by $G^{2/3}$. This contraction is not squeezing below vacuum.

Checks. In GPT 6 Pro's finite-volume covariance code (family gpt) the fitted powers over $12 \leq \tau \leq 26$ are 0.32461 and 1.69203; fits between equal phases of the oscillation give 0.33349 and 1.66676 [P]. In gpt-e's clean-context integration (gpt) the compensated $\ell = 2$ variances $G^{-1/3}N_{\rm out}\mathrm{Var}\, n_2$ and $G^{-5/3}\varepsilon_0^2 N_{\rm out}\mathrm{Var}\, p_2$ stay within [1.11, 2.68] and [0.186, 0.447] for $12 \leq \tau \leq 30$ [E] [Fig. 1(a)]. The full NLSE, with all dispersive terms, a rounded asymptotically chirped seed, vacuum and both ASE quadratures, gives an $\ell = 2$ decay rate 0.333825 and log-gain frequency 0.470401 (predicted 1/3 and 0.471405) over $1 \leq \tau \leq 6$ at $\varepsilon_0 = 0.05$; doubling the grid and halving the step change the rate by 2×10^{-10} [P]. gpt-a's full-field rates are 0.33316, 0.33375 and 0.33259 for $\ell = 2, 3, 4$ [A]. No full-NLSE noise computation exists for $\tau \geq 12$.

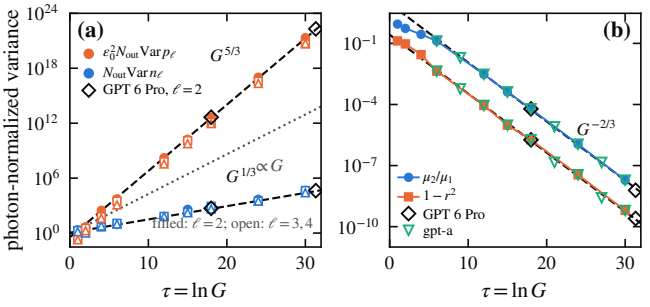


FIG. 1. Shape-mode noise in the reduced model ($\Gamma = \sqrt{18}$, $n_{sp} = 1$, seed on the asymptote). (a) Photon-normalized intensity (blue) and phase (orange) variances of the orthonormal Legendre components $\ell = 2$ (filled), 3, 4 (open); dashed: $G^{1/3}$, $G^{5/3}$; dotted: $\propto G$, a linear amplifier. (b) Eigenvalue ratio μ_2/μ_1 of the five-component spectral covariance and phase–energy deficit $1-r^2$; dashed: $G^{-2/3}$. Coloured symbols: gpt-e's clean-context integration, run `2026-10-19-gpt-e-similariton`. Diamonds: GPT 6 Pro at equal phases of the oscillation, `2026-10-18-pi-chatgpt-independent-verification`. Triangles: gpt-a, `2026-10-11-gpt-a-highlights-audit-c2`.

IV. SPECTRAL RANK ONE AND ENERGY–PHASE LOCKING

A. Single-shot spectrum

The output is strongly chirped, so a dispersive Fourier transform maps time to frequency, $\omega \simeq \kappa T$ with $\kappa = \dot{T}_p/(\beta_2 T_p)$. Let $s = \delta S/S_0$, with S_0 the peak of the mean spectrum and Legendre components s_ℓ over the spectral core, and let $\mu_1 \geq \mu_2$ be the largest eigenvalues of the covariance of $(s_0, \ldots, s_4)$ in the orthonormal $L^2[-1,1]$ metric.

Statement (R-0011b). On the asymptote $\mu_2/\mu_1 = \Theta(G^{-2/3})$, for this five-component projection.

Crux. Stationary phase gives $s_\ell = n_\ell - \ell(\ell+1)\varepsilon_0 p_\ell/(aa')$, which on the asymptote is $s_\ell = n_\ell + 3n_\ell'$. The energy component $s_0 = \frac{2}{3}\delta N/N$ is fixed once injected, while $s_{\ell\geq 1} = O(G^{-1/3})$; for $\ell = 1$ the secular term cancels, since $n_1 = (c_1 + c_2\tau)e^{-\tau/3}$ gives $s_1 = 3c_2 e^{-\tau/3}$. The covariance therefore has the block form

$$C_s = \begin{pmatrix} c_0 & G^{-1/3}b^{\mathsf{T}} \\ G^{-1/3}b & G^{-2/3}D \end{pmatrix}, \quad (5)$$

and if $c_0 > 0$ and the Schur complement $D - bb^{\mathsf{T}}/c_0$ stays positive and bounded, $\mu_1 = c_0 + O(G^{-2/3})$ while $\mu_2 = \Theta(G^{-2/3})$. The single-shot spectral noise collapses onto the energy fluctuation. This is not a statement about the full field, nor one that the quantum state becomes rank one.

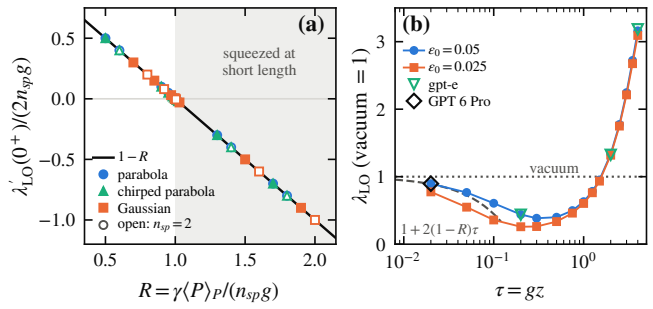


FIG. 2. Squeezing of the matched mode at short length despite full ASE. (a) Initial slope of the matched-mode minimum variance, divided by $2n_{sp}g$, against R; line: Eq. (10). Smoothed parabolas (unchirped and chirped) and a Gaussian; open symbols: held-out $n_{sp} = 2$. (b) $\lambda_{\rm LO}$ against τ in the full field ($\Gamma = \sqrt{18}$, $n_{sp} = 1$; parabola smoothed at $\delta = 0.05$, with the asymptotic chirp), vacuum and both reservoir quadratures included; dashed: first order, $1 + 2(1 - R)\tau$ with $R = 3.76$ for $\varepsilon_0 = 0.05$. Lines: run `2026-10-19-claude-interactive.13-similariton-lo-verification`. Triangles: gpt-e, chirped unsmoothed launch, $\varepsilon_0 = 0.05$, `2026-10-19-gpt-e-similariton`. Diamond: GPT 6 Pro's counterexample on its own launch, `2026-10-18-pi-chatgpt-independent-verification`.

B. Matched local oscillator

Statement (R-0011c, R-0014i). In the reduced model on the asymptote,

$$C_{AA} = 2n_{sp}G - (2n_{sp} - 1), \tag{6}$$

$$C_{PP} = \tfrac{32n_{sp}}{9}\, G\, \Phi_{NL}^2 \left[1 + O(G^{-1/3})\right] \propto G^{7/3}/\varepsilon_0^2, \tag{7}$$

$$1 - r^2 = \Theta(G^{-2/3}). \tag{8}$$

Crux. Equation (6) is exact in the linearized theory, for any pulse and dispersion: Kerr and dispersion conserve photon number, so $X_A = \delta N/\sqrt{N}$ sees only the gain and its reservoir, a linear amplifier. The phase follows from the $\ell = 0$ block, $p_0' = \Phi_{NL}' n_0$, where $n_0 = \frac{2}{3}\delta N/N$ changes only by ASE kicks dW_0 of variance $\propto e^{-\tau} d\tau/N_0$. Integrating by parts,

$$p_0 = \Phi_{NL} n_0 + R_0, \qquad R_0 = -\int_0^\tau \Phi_{NL}\, dW_0. \tag{9}$$

The first term has variance $\Phi_{NL}^2 \mathrm{Var}\, n_0 \to \frac{8n_{sp}}{9}\Phi_{NL}^2/N_0$, of order $G^{4/3}/(\varepsilon_0^2 N_0)$; the residual has $\mathrm{Var}\, R_0 \propto \varepsilon_0^{-2}\int^\tau e^{4\tau'/3} e^{-\tau'} d\tau'/N_0 = O(G^{1/3}/\varepsilon_0^2 N_0)$. With $C_{PP} = 4N_{\rm out}\mathrm{Var}\,\bar{p}$ this gives Eq. (7); the leading correction is the p_0–p_2 cross-covariance, $\Theta(G)$ against $\Theta(G^{4/3})$. The part of $\bar{p}$ not linear in the energy is dominated by the shape phase $p_2/5$, whose conditional variance is $\Theta(G^{2/3}/\varepsilon_0^2 N_0)$ by Sec. III; dividing by the total gives Eq. (8), independent of ε_0. The phase noise is the Gordon–Mollenauer conversion of energy noise [14], accumulated over a nonlinear phase that itself grows as $G^{2/3}$. The original claim's $C_{PP} \propto G/\varepsilon_0^2$ contradicted its own $\ell = 0$ formula and is superseded.

Checks. Reduced model: fitted powers −0.66551 (spectral ratio) and −0.65351 (deficit) over $12 \le \tau \le 26$, and −0.66734 for the deficit between equal phases [P]. Three gpt-family implementations agree, for example $\mu_2/\mu_1 = 3.03 \times 10^{-3}$ and $1 - r^2 = 8.96 \times 10^{-5}$ at $\tau = 12$ [P, A, E] [Fig. 1(b)]. The fitted power of $\varepsilon_0^2 C_{PP}$ is 2.3335 [P] and, in claude-interactive.13's code (family claude), 2.33350 over $12 \le \tau \le 26$ and 2.333333 over 26–40; at the held-out $\Gamma = 1$ it is 2.33175 and 2.333314 [V]. The error of the prefactor in Eq. (7) falls as $G^{-0.334}$ ($\Gamma = \sqrt{18}$) and $G^{-0.332}$ ($\Gamma = 1$), against the predicted $G^{-1/3}$ [V] [Fig. 3(a)]; at $\tau = 2$, 4, 6 the ratio is 1.098, 1.106, 1.024 ($\Gamma = \sqrt{18}$) and 0.766, 0.855, 0.935 ($\Gamma = 1$) [V]. Equation (6) holds to 7×10^{-15} in the reduced model and to order 10^{-11} in the full field [V]. The full NLSE with complete noise at $\tau = 4$ gives spectral ratios 0.31249 and 0.29547 and deficits 0.02920 and 0.02841 at $\varepsilon_0 = 0.05$ and 0.025, moving towards the leading-order values of about 0.278 and 0.02804; no error power in ε_0 was isolated [P].

V. SQUEEZING DESPITE AMPLIFIER NOISE, AND ITS LIMIT

A. Short length

Statement (R-0014ii). For any smooth coherent pulse and any dispersion, the smaller eigenvalue of $C_{\rm LO}$ obeys

$$\left.\frac{d\lambda_{\rm LO}}{dz}\right|_{0^+} = 2n_{sp}g - 2\gamma\frac{\int P^2 dt}{\int P\, dt} \equiv 2n_{sp}g\,(1 - R). \tag{10}$$

The matched mode is squeezed at short length if $R > 1$, and not if $R < 1$. The thresholds are $\gamma P_0 > \frac{5}{4}n_{sp}g$ for a parabola ($\int P^2/\int P = \frac{4}{5}P_0$) and $\gamma P_0 > \sqrt{2}\, n_{sp}g$ for a Gaussian $P_0 e^{-t^2/T_0^2}$.

Crux. Expand the linearized covariance equation, $dV/dz = AV + VA^{\mathsf{T}} + D$, at $z = 0^+$, where $V = I$. Gain drift and ASE add $gI + (2n_{sp} - 1)gI = 2n_{sp}gI$. Dispersion and phase rotations are passive, $A + A^{\mathsf{T}} = 0$, and do nothing to the isotropic vacuum. Kerr, in the local frame of the mean field, is the shear $dy/dz = 2\gamma P(t)\, x$ of the phase quadrature y by the amplitude quadrature x, whose symmetric part adds $2\gamma P(t)$ to the amplitude–phase covariance at each t. The matched mode weights time slots with $|\bar{A}|^2/N = P/\int P$, so on $(X_A, P_{\rm LO})$

$$\left.\frac{dC_{\rm LO}}{dz}\right|_{0^+} = 2n_{sp}g\, I + 2\gamma\langle P\rangle_P\, \sigma_x, \qquad \langle P\rangle_P = \frac{\int P^2}{\int P}, \tag{11}$$

whose smaller eigenvalue is Eq. (10). Phase-insensitive gain adds noise to every quadrature at rate $2n_{sp}g$; the Kerr shear removes it from one quadrature at rate $2\gamma\langle P\rangle_P$. Caves' theorem [12] requires added noise of phase-insensitive linear amplifiers; linearized, the Kerr amplifier is phase sensitive, and a phase-sensitive map may leave one quadrature below vacuum. At $R = 1$ the second order decides. Without dispersion the $(gz)^2$ coefficient is $2\mathrm{Var}_P(\gamma P/g) - n_{sp}$ (Appendix E), negative for both shapes at $n_{sp} = 1$, 2, so the mode is squeezed there too and the original "if and only if" fails on the boundary.

Checks. claude-interactive.13's full-field code (claude), with both vacuum and both reservoir quadratures and an exact discrete adjoint, gives the initial slope $2n_{sp}g(1-R)$ to 7×10^{-7} and its root within 3.3×10^{-9} of $R=1$; the reservoir restores the commutator to 4×10^{-13} [V] [Fig. 2(a)]. The held-out $n_{sp}=2$ thresholds in $\gamma P_0/g$ are 2.828427 for the Gaussian ($2\sqrt{2}$) and 2.50699, 2.50030, 2.500076 for parabolas smoothed at $\delta=0.05$, 0.01, 0.005 (5/2) [V]. GPT 6 Pro's counterexample to the original sentence "with ASE there is no squeezing", $\lambda_{\rm LO}=0.8964864655$ at $\tau=0.02$ ($\varepsilon_0=0.05$, $\Gamma=\sqrt{18}$, $n_{sp}=1$, $R\approx3.8$) [P], was reproduced on a second platform [Q] and with the verifier's code, 0.8964864646–53 [V]; gpt-e's clean-context audit found 0.436717 at $\tau=0.2$ [E] [Fig. 2(b)]. With these parameters the mode stays squeezed up to $\tau\approx1.6$ [V]. At $R=1$ the full-field second-order coefficients are −0.85, −1.41 (smoothed parabola, $n_{sp}=1$, 2) and −0.69, −0.76 (Gaussian) [V]. Truncated Wigner simulation at $N_0=10^6$ reproduces the linearized squeezing within one standard error, $\lambda_{\rm LO}=0.4729\pm0.0013$ against 0.4729 (Gaussian, $n_{sp}=1$, $\tau=0.3$) [V]. At $N_0\le10^4$ it gives less squeezing (0.488–0.492 at 10^4, 0.63–0.69 at 10^3), not separated from the method's validity limit.

B. Large gain

Statement (R-0014iii). In the reduced asymptotic model the matched mode is not squeezed:

$$\lambda_{\min}(C_{\rm LO})=\frac{\det C_{\rm LO}}{\lambda_{\max}}\simeq C_{AA}(1-r^2)=\Theta(G^{1/3}). \qquad (12)$$

Crux. $\det C_{\rm LO}=C_{AA}C_{PP}(1-r^2)$ and $\lambda_{\max}\simeq C_{PP}$, so Eqs. (6) and (8) give $\Theta(G)\,\Theta(G^{-2/3})$, independent of ε_0. Energy noise injected early is amplified as G; the phase locks to it, but only up to the shape phase, a fraction $G^{-2/3}$. For this mode to stay squeezed the locking would have to improve faster than G^{-1}. The statement concerns this mode, not every detection mode.

Checks. $G^{-1/3}\lambda_{\min}$ stays in [0.338, 0.633] for $12\le\tau\le26$ and [0.338, 0.625] out to $\tau=80$ ($\Gamma=\sqrt{18}$, $n_{sp}=1$), and in [0.247, 0.797] at $\Gamma=1$ [V]; gpt-e's reduced covariances agree at the sampled gains [E] [Fig. 3(b)]. In the full field at $\tau=4$ and $\varepsilon_0=0.05$, $\lambda_{\rm LO}=3.1592357$ against the proposer's 3.159; it lies 4.2% above the reduced model at $\varepsilon_0=0.05$ and 1.3% above at 0.025 [V].

VI. STATUS AND LIMITS

Assumptions. All statements assume a scalar field in one polarization and one spatial mode, instantaneous Kerr response (no Raman), second-order dispersion only, flat unsaturated gain with a complete white reservoir, no loss, coherent input, and noise linearized about the bright field. R-0011 and parts (i) and (iii) of R-0014 also assume the leading order in ε_0 with regular, bounded-flux edges, and keep only the intensity quadrature of the input and ASE noise: the phase quadrature

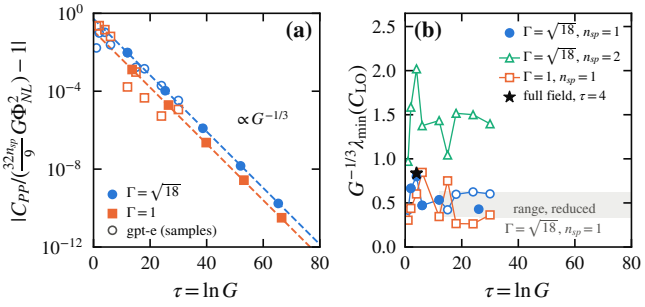


FIG. 3. The matched LO at large gain, reduced model. (a) Error of the prefactor in Eq. (7): filled, its envelope at equal phases of the oscillation, `2026-10-19-claude-interactive.13-similariton-lo-verification` (seed on the asymptote for $\Gamma=\sqrt{18}$, unchirped for $\Gamma=1$; $n_{sp}=1$); dashed: $\propto G^{-1/3}$; open: gpt-e's covariances at sampled gains, `2026-10-19-gpt-e-similariton`. (b) Compensated minimum variance of the matched mode, Eq. (12). Filled circles and grey band: values and range of the first run; open symbols: gpt-e. Stars: full field at $\tau=4$, $\varepsilon_0=0.05$ (filled: proposer's launch, first run; open: gpt-e).

enters with weight $1/(1-\xi^2)$, singular at the edges, and is dropped, so the reduced covariance is not a complete quantum state. The full-field checks relax the hydrodynamic order, the sharp edge and the dropped phase noise; none relaxes the scalar model or the flat gain.

Brightness and bandwidth. $\Phi_{NL}^2/N_0\ll1$ fails at high gain: at $\varepsilon_0=0.05$, $\Phi_{NL}^2=4.44\times10^8$ at $\tau=12$ and 5.68×10^{16} at $\tau=26$ [V], while a 500 pJ seed at 1.03 μm carries $N_0=2.6\times10^9$ photons [S]; with $N_0=10^4$ the rms phase is 199 rad at $\tau=12$ [P]. Beyond $\tau\approx12$ the large-gain statements are therefore statements about the linearized model. The spectral edge grows as $G^{1/3}$, so at fixed gain bandwidth Ω_g the asymptotic window closes once $gL(\omega_{\rm edge}/\Omega_g)^2$ reaches about 0.05; the asymptotics are an intermediate regime.

What failed. Besides the two refuted sentences, the original claim's uniform edge remainder $O(\varepsilon_0^2\ln\varepsilon_0^{-1})$ is not verified: for a sharp-edged seed the raw phase projections diverge logarithmically [P], and sharp-edge full-field moments depend on the resolution, while smoothing the edge restores convergence in the tested case [E]. Some finite-gain covariance tables of that claim fail full-field checks: at the held-out $\Gamma=1$, $gL=2$, the spectral ratio is 0.385841 and 0.385432 at $\varepsilon_0=0.05$ and 0.025, against the tabulated 0.4257 [A]. The rate at which the full field approaches the reduced model in ε_0 is not established.

Context, not verified. Three claims on the same problem have only the proposer's checks. C-20261008-claude-a-01 gives exact moment equations for the energy, centroid-frequency and timing noise, driven by the pulse's own time–frequency covariance; on the similariton they predict a centroid-frequency noise figure $6n_{sp}-2$ and a compressed-pulse timing variance growing as $(gL)^2$. C-20261008-gpt-a-01 gives, within the parabolic family, a Lyapunov function $V=\dot{s}^2/2+s^2/18+1/(9s)$ with $\dot{V}=-\frac{2}{3}\dot{s}^2$ for the scaled width s, and a conditional nonlinear attraction bound; its width rates $(-1\pm i\sqrt{2})/3$ are the $\ell=2$ rates of Eq. (4). C-20261008-gpt-

TABLE I. Status of each statement (October 19, 2026). "Promising" is the repository's tier for a result verified by simulation in a second model family and approved by the PI; "formalized" would also need a kernel-checked Lean proof. Gates G1–G8 are those of the repository's promotion rules: G4 is cross-family simulation, G5 a clean-context adversarial audit, G6 a novelty clearance, G7 a reproduction from committed files, G8 formalization. Families in parentheses. Claim ids: a-02 is C-20261008-claude-a-02, pi-02 is C-20261019-pi-chatgpt-02.

Statement	Tier and gates	Proposer; verifiers; auditor	Regime	Narrowed, failed or open
Shape-mode envelopes, Eq. (2) [R-0011a; a-02]	promising; G1–G5 met, G6–G8 open	claude-a (claude); GPT 6 Pro and gpt-a (gpt); clean-context audit gpt-e (gpt), pass with reduced regime	leading order in ε_0 (full field $\varepsilon_0 \le 0.1$); $12 \le \tau \le 26$; $\ell = 2$–4, $\ell\varepsilon_{\rm eff} \le 0.2$; $0.5 \le \Gamma \le 10$ (tested $\sqrt{18}$, 1); $\Phi_{NL}^2/N_0 \ll 1$	envelopes, not limits; uniform edge remainder unverified; no full-field noise at $\tau \ge 12$
Spectral ratio $\mu_2/\mu_1 = \Theta(G^{-2/3})$ [R-0011b; a-02]	as above	as above	as above, $\ell \le 4$, $aa'/\varepsilon_0 \ge 20$, $gL(\omega_{\rm edge}/\Omega_g)^2 \le 0.05$, $n_{sp}M/N_0 \ll 1$	five-component projection only; finite-gain table entries fail ($\Gamma = 1$, $gL = 2$)
Deficit $1 - r^2 = \Theta(G^{-2/3})$ [R-0011c; a-02]	as above	as above	as row 1	full-field large-gain exponent not tested
Short-length criterion, Eq. (10) [R-0014ii; pi-02]	promising; G1–G4 met, G5–G8 open	GPT 6 Pro (gpt); claude-interactive.13 (claude); no audit. Supporting: gpt-e (gpt, proposer's family)	any smooth coherent pulse and dispersion, $z \to 0^+$; $R \ne 1$; $1 \le n_{sp} \le 2$ tested; $\Phi_{NL}^2/N_0 \ll 1$	"if and only if" superseded (fails at $R = 1$); finite-length threshold below $R = 1$ unverified
C_{AA}, C_{PP}, Eqs. (6), (7) [R-0014i; pi-02]	as above	as above (without gpt-e)	reduced model; $12 \le \tau \le 26$ (tested to 80); $0.5 \le \Gamma \le 10$; $\varepsilon_0 \le 0.1$; $\Phi_{NL}^2/N_0 \ll 1$	correction $O(G^{-1/3})$; ratio 0.77–1.11 at $\tau = 2$–6
$\lambda_{\rm min} = \Theta(G^{1/3})$, Eq. (12) [R-0014iii; pi-02]	as above	as above	as above	this mode only
$C_{PP} \propto G/\varepsilon_0^2$; "with ASE there is no squeezing" [a-02, original]	refuted	claude-a (claude); refuted by GPT 6 Pro (gpt)	—	replaced by Eqs. (7), (10), (12)
Collective-coordinate moments [C-20261008-claude-a-01]; attraction [C-20261008-gpt-a-01, -02]	not verified (proposer's checks only)	claude-a (claude); gpt-a (gpt)	—	context only (Sec. VI)

a-02 shows that strong global attraction is false, because odd inputs stay orthogonal to the even similariton.

VII. WHAT TO MEASURE

Short-length squeezing is the most accessible test. $R > 1$ is easy to reach: the energetic parabolic seed of C-20261008-claude-a-01 (3 ps, 500 pJ, $\beta_2 = 25$ ps^2/km, $\gamma = 5$ W^{-1}km^{-1}, $g = 0.2$ m^{-1}) has $\gamma P_0/g = 3.1$ and $R = 2.5/n_{sp}$. Homodyne detection of the output with an LO derived from the output mean field should show sub-shot-noise variance over a short gain length when $R > 1$, with initial slope $2n_{sp}g(1 - R)$; lowering γP_0 or raising n_{sp} across $R = 1$ should turn the slope positive. Loss and detection efficiency η map $\lambda \mapsto 1 - \eta + \eta\lambda$. No sub-shot-noise matched-mode variance at R clearly above 1, after this correction, would falsify Eq. (10).

Deterministic shape transfer is accessible now: a calibrated $\ell = 2$ modulation of the seed should decay as $e^{-\tau/3}$ and oscillate at $\sqrt{2}/3$ in $\ln G$, read by FROG or from the chirped spectrum. This tests Eq. (3), not the noise, and overlaps the published width dynamics.

Spectral covariance by dispersive Fourier transform: μ_2/μ_1 should fall as $G^{-2/3}$ once averaged over a covariance period $\pi/\nu_2 = 3\pi/\sqrt{2}$ in $\ln G$; in the reduced model it is 3.0×10^{-3} at $\ln G = 12$ [P]. This needs more than 50 dB of flat gain, a bright seed, and technical noise below the shot-noise-referenced shape fluctuations. A ratio that does not fall with gain, or falls with a clearly different period-averaged power, falsifies the statement. Spectrally sliced intensity-noise measurements in gain-managed amplifiers [9] are the nearest existing experiments. The same holds for $1 - r^2$ from simultaneous energy and phase measurements against the mean field, and for the $G^{7/3}$ phase growth, within the brightness limit.

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Appendix A: Hydrodynamic blocks with gain

Normalization. With $A = \sqrt{P_0\rho}\, e^{i\phi}$, the dispersionless limit of Eq. (1) gives $\phi_\zeta = \frac{\varepsilon_0}{2}\phi_x^2 + \rho/\varepsilon_0$ and $\rho_\zeta + (\rho v)_x = \Gamma\rho$ with $v = -\varepsilon_0\phi_x$; differentiating the first gives $v_\zeta + v v_x = -\rho_x$. The parabola of Sec. II has energy $\int \bar\rho\, dx = \frac{4}{3}\rho_{\rm pk} a = \frac{4}{3}e^{\Gamma\zeta}$, so $\rho_{\rm pk} = e^{\Gamma\zeta}/a$, and the momentum equation fixes $a'' = 2\rho_{\rm pk}/a$.

Linearization. Put $\rho = \bar\rho + \rho_{\rm pk} n$ and $\phi = \bar\phi + p$, and change to (ξ, ζ) with $\partial_\zeta|_x = \partial_\zeta|_\xi - (a'/a)\xi\partial_\xi$, $\partial_x = a^{-1}\partial_\xi$. The phase equation becomes $\partial_\zeta|_\xi p = \rho_{\rm pk} n/\varepsilon_0 = (e^{\Gamma\zeta}/a)\, n/\varepsilon_0$: the phase is advected with the background and driven by the intensity. In the continuity equation the terms $\partial_\zeta(\rho_{\rm pk} n) + \partial_x(\bar v \rho_{\rm pk} n)$ reduce to $\rho_{\rm pk}(\partial_\zeta|_\xi n + \Gamma n)$, because $\rho'_{\rm pk}/\rho_{\rm pk} = \Gamma - a'/a$; the Γn term cancels the gain. The remaining flux term is $-\partial_x(\bar\rho\, \varepsilon_0 \partial_x p) = -(\varepsilon_0\rho_{\rm pk}/a^2)\, \partial_\xi[(1-\xi^2)\partial_\xi p]$. Hence $\partial_\zeta|_\xi n = (\varepsilon_0/a^2)\, \partial_\xi[(1-\xi^2)\partial_\xi p]$. Regular solutions with bounded flux at $\xi = \pm 1$ expand in Legendre polynomials, and Eq. (3) follows for any $a(\zeta)$. Its block map has determinant 1 (it is Hamiltonian in $(p_\ell, n_\ell/\varepsilon_0)$).

Asymptote. Eliminating p_ℓ from Eq. (3) gives $q'' + 2(a'/a)q' + \ell(\ell+1)(a''/2a)q = 0$ for $q = n_\ell/\varepsilon_0$ (derivatives in ζ); with $a = A_* e^{\Gamma\zeta/3}$, $A_*^3 = 18/\Gamma^2$ and $d/d\zeta = \Gamma\, d/d\tau$ this is Eq. (4). In $u = e^{\tau/3} n_\ell$, $w = \varepsilon_0 e^{-\tau/3} p_\ell$,

$$\frac{d}{d\tau}\begin{pmatrix} u \\ w \end{pmatrix} = B_\ell \begin{pmatrix} u \\ w \end{pmatrix}, \quad B_\ell = \begin{pmatrix} \frac{1}{3} & -\frac{\ell(\ell+1)}{\Gamma A_*^2} \\ \frac{1}{\Gamma A_*} & -\frac{1}{3} \end{pmatrix}, \tag{A1}$$

and $B_\ell^2 = [\frac{1}{9} - \ell(\ell+1)/(\Gamma^2 A_*^3)]I = -\nu_\ell^2 I$, so $e^{B_\ell\tau} = \cos\nu_\ell\tau\, I + \nu_\ell^{-1}\sin\nu_\ell\tau\, B_\ell$ is bounded. For $\ell = 0$, n_0 is constant and p_0 grows with Φ_{NL}; for $\ell = 1$, $\nu_1 = 0$ and $n_1 = (c_1 + c_2\tau)e^{-\tau/3}$.

Two-sided bounds. The intensity source in (u, w) is $\propto e^{-\tau/3}$ (Appendix B), and a regulated phase source would be $\propto \varepsilon_0^2 e^{-5\tau/3}$, so the interaction-picture integrals converge. Intensity diffusion plus the drift coupling makes each block controllable, so the limiting covariance is positive definite and the rotation keeps it between two positive multiples of the identity. This is GPT 6 Pro's argument [P], for a fixed finite truncation and a fixed edge regulator; the covariance oscillates with period π/ν_ℓ in $\ln G$ (6.66 for $\ell = 2$).

Appendix B: Noise sources and photon normalization

For a coherent state, symmetrically ordered, $\langle \delta A\, \delta A'^* \rangle = \frac{1}{2}\delta(T-T')$, so $\delta|A|^2 = 2{\rm Re}(\bar A^*\delta A)$ has $\langle \delta\rho\, \delta\rho'\rangle = \bar\rho\, \delta(T-T')$. Projecting $n = \delta\rho/\rho_{\rm pk}$ onto $P_\ell(\xi)$ with $n_\ell = \frac{2\ell+1}{2}\int n P_\ell\, d\xi$ and using $N = \frac{4}{3}\rho_{\rm pk} T_p$,

$$\mathrm{Cov}(n_\ell, n_m) = \frac{(2\ell+1)(2m+1)}{3N}\, B_{\ell m}, \tag{B1}$$

with $B_{\ell m} = \int_{-1}^{1}(1-\xi^2) P_\ell P_m\, d\xi$. This metric is not diagonal ($B_{02} = -\frac{4}{15}$), which couples n_0 and n_2 through the noise only. Distributed ASE adds $(2n_{sp} - 1)$ times the same covariance per unit τ, evaluated with $N(\tau) = N_0 e^\tau$; for $\ell = 0$ this reproduces the linear-amplifier energy noise ${\rm Var}(\delta N)/N^2 = [1 + (2n_{sp}-1)(1 - G^{-1})]/N_0$. The phase quadrature, $\delta\arg A = {\rm Im}(\bar A^*\delta A)/|\bar A|^2$, has variance density $\propto 1/\bar\rho \propto 1/(1-\xi^2)$, singular at the edges; at leading order it is dropped (assumption of Sec. VI). Multiplying ${\rm Var}\, n_\ell = O(G^{-2/3}/N_0)$ and ${\rm Var}\, p_\ell = O(G^{2/3}/\varepsilon_0^2 N_0)$ by $N_{\rm out} = G N_0$ gives the photon-normalized envelopes $G^{1/3}$ and $G^{5/3}$; their rms values grow as $G^{1/6}$ and $G^{5/6}$.

Appendix C: Spectral map and Schur complement

For a strongly chirped output, $aa'/\varepsilon_0 \geq 20$, stationary phase maps T to $\omega = \kappa T$. A density fluctuation moves spectral weight directly, and a phase fluctuation shifts the local frequency by $-\partial_T p$, which the chirp maps to a displacement of the time-to-frequency map; to leading order $s_\ell = n_\ell - \ell(\ell+1)\varepsilon_0 p_\ell/(aa')$. On the asymptote $aa' = \frac{\Gamma}{3}A_*^2 e^{2\tau/3}$ and $n'_\ell = -\varepsilon_0\ell(\ell+1)p_\ell/(\Gamma A_*^2 e^{2\tau/3})$, so $s_\ell = n_\ell + 3n'_\ell$. The energy mode is uniform in the spectral core because a stronger pulse is both taller and wider. For the block matrix (5) with $c_0 > 0$, the eigenvalues are $\mu_1 = c_0 + O(G^{-2/3})$ and, to leading order, $G^{-2/3}$ times the eigenvalues of $D - bb^{\rm T}/c_0$, so $\mu_2/\mu_1 = \Theta(G^{-2/3})$ whenever that Schur complement is uniformly positive definite and bounded. That condition is assumed, not proved; the reduced-model numerics of Fig. 1(b), from three implementations, are consistent with it.

Appendix D: Matched local oscillator

Number quadrature. $X_A = 2{\rm Re}\langle \bar A/\sqrt{N}, \delta A\rangle = \delta N/\sqrt{N}$. Since $\int |A|^2$ is conserved by Kerr and dispersion, $d\,\delta N/dz = g\,\delta N + 2{\rm Re}\langle \bar A, F\rangle$, and the symmetrized noise gives $d\,{\rm Var}(\delta N)/dz = 2g\,{\rm Var}(\delta N) + (2n_{sp}-1)gN$. With $C_{AA} = {\rm Var}(\delta N)/N$ and $C_{AA}(0) = 1$ this integrates to $C_{AA} = G + (2n_{sp}-1)(G-1)$, Eq. (6), for any pulse shape, dispersion and nonlinearity.

Phase quadrature. With $\theta = \varepsilon_0\Phi_{NL}$, the $\ell = 0$ block gives $p_0(L) = \varepsilon_0^{-1}[\theta_L n_0(0) + \int_0^L (\theta_L - \theta)\, dW_0]$, so

$$\varepsilon_0^2 N_0 {\rm Var}\, p_0 = \tfrac{4}{9}\Big[\theta_L^2 + (2n_{sp}-1)\Gamma\int_0^{\zeta_L} e^{-\Gamma\zeta}(\theta_L - \theta)^2 d\zeta\Big], \tag{D1}$$

the formula already on the original claim card. For large G the integral tends to θ_L^2, so ${\rm Var}\, p_0 \to \frac{8n_{sp}}{9}\Phi_{NL}^2/N_0$, and $C_{PP} = 4N_0 G\,{\rm Var}\,\bar p$ gives Eq. (7). The cross term $-\frac{2}{5}{\rm Cov}(p_0, p_2)$ is nonzero because $B_{02} \neq 0$; the $\ell = 2$ response to an early kick grows as $e^{\tau/3}\sin[\nu_2(\tau - \tau')]$, so the cross term is $\Theta(G)$ against $\Theta(G^{4/3})$, a relative correction $O(G^{-1/3})$ that oscillates with period $2\pi/\nu_2 = 13.3$ in $\ln G$. This is the rate measured in Fig. 3(a).

Deficit and minimum. Regressing $\bar p = \Phi_{NL} n_0 + R_0 - p_2/5$ on n_0 leaves $R_0 - p_2/5$ minus its projection on n_0. ${\rm Var}\, R_0 = O(G^{1/3}/\varepsilon_0^2 N_0)$ and the conditional variance of p_2 is $\Theta(G^{2/3}/\varepsilon_0^2 N_0)$ when the $\ell = 2$ noise is not fully correlated with the $\ell = 0$ noise, which the metric B ensures. With

$\mathrm{Var}\,\bar{p} = \Theta(G^{4/3}/\varepsilon_0^2 N_0)$ this gives $1 - r^2 = \Theta(G^{-2/3})$. For C_{LO} with $C_{PP} \gg C_{AA}$, $\lambda_{\min} = \det C/\lambda_{\max} = C_{AA}(1-r^2)[1 + O(C_{AA}/C_{PP})]$, Eq. (12).

Appendix E: Short-length expansion to second order

Without dispersion each time slot evolves separately. In the local frame of the mean field, the slot covariance $V(t,z)$ of (x,y) obeys $dV/dz = gV + KV + VK^{\mathsf{T}} + (2n_{sp}-1)gI$, with $K = \begin{pmatrix} 0 & 0 \\ k & 0 \end{pmatrix}$ and $k = 2\gamma P(t) e^{gz}$, and the matched-mode covariance is $C = \int (P/\iint P)\, V\, dt$. With $V(0) = I$,

$$V'(0) = 2n_{sp} g I + k\sigma_x, \tag{E1}$$

$$V''(0) = 2n_{sp} g^2 I + 2g(1+n_{sp}) k \sigma_x + 2k^2 e_{yy}, \tag{E2}$$

where $e_{yy} = \mathrm{diag}(0,1)$. Averaging over $P/\iint P$ replaces k by $\bar{k} = 2\gamma\langle P\rangle_P$ and k^2 by $\overline{k^2}$. The smaller eigenvalue is $\lambda_{\mathrm{LO}} = 1 + z(2n_{sp}g - \bar{k}) + \frac{z^2}{2} v^{\mathsf{T}} C'' v + O(z^3)$ with $v = (1,-1)/\sqrt{2}$, that is

$$\lambda_{\mathrm{LO}} = 1 + z(2n_{sp}g - \bar{k}) + z^2\left[n_{sp} g^2 - (1+n_{sp}) g\bar{k} + \tfrac{1}{2}\overline{k^2}\right] + O(z^3). \tag{E3}$$

At $R = 1$, $\bar{k} = 2n_{sp}g$ and $\overline{k^2} = 4g^2[\mathrm{Var}_P(\gamma P/g) + n_{sp}^2]$, so the bracket is $g^2[2\mathrm{Var}_P(\gamma P/g) - n_{sp}]$. For a sharp parabola $\mathrm{Var}_P(\gamma P/g) = n_{sp}^2/14$ at $R = 1$, so the coefficient is $n_{sp}^2/7 - n_{sp}$: $-6/7$ and $-10/7$ for $n_{sp} = 1, 2$. For a Gaussian it is $n_{sp}^2(4/\sqrt{3} - 2) - n_{sp}$: -0.6906 and -0.7624. These closed forms are derived here; they match the record's full-field values for the Gaussian and, for the parabola, lie close to the values for a parabola smoothed at $\delta = 0.05$ (Sec. V). For the parabola the coefficient is negative only for $n_{sp} < 7$; this has not been checked by another family.

Appendix F: Numerical methods of the checks

GPT 6 Pro [P] wrote a finite-volume covariance integrator for the reduced model (32–1024 cells; at $\tau = 26$ the compensated variances change by 2.2×10^{-5} between 512 and 1024 cells) and a lens-coordinate full NLSE with exact tangent and adjoint and both vacuum and both ASE quadratures (commutator residual below 6.3×10^{-11} at $\tau = 4$). It found and corrected a readout defect in its own code (a midpoint readout leaked $O(h^2)$ of the energy mode into the shape modes at high gain); the defective data are kept as a diagnostic. It read the claim's derivation notes, so it counts as a verification, not a clean-context audit. *gpt-e* [E] wrote, in clean context, a zero-flux finite-volume operator with an explicitly orthonormal output projection (256 cells, 8 modes) and a Fourier full-field ASE contraction, with a spot check by the unchanged pinned solver's discrete adjoint. *gpt-a* [A] wrote an independent reduced modal model and full-field rate fits. *claude-interactive.13* [V] derived (ii) and the $O(G^{-1/3})$ rate before reading the proposer's material, then wrote a reduced Legendre covariance code (tolerances 10^{-9} to 10^{-13} change the power by 10^{-14}), a full-field linearized code with an exact discrete adjoint ($N = 2048$–8192, windows 8–12; steps $\times 4$ change $\lambda_{\mathrm{LO}}(\tau = 0.02)$ by at most 10^{-9}) and a truncated-Wigner code (2×10^5 trajectories). No repository solver was used for the results; the codes are independent of the proposers'.

Appendix G: Provenance of the numbers

Table II lists every number quoted in the text with its run and the agent that produced it; run ids are folders under `runs/` in the project repository, abbreviated as in Sec. II. Values marked "fig." were computed by `make_figures.py` from the committed numbers of the cited run; so is $R = 3.76$ in Fig. 2(b), from $\langle P\rangle_P/P_0 = 0.7978$ of the smoothed parabola in run [V]. Closed-form evaluations ($R \approx 3.8$, $\gamma P_0/g = 3.1$, $R = 2.5/n_{sp}$, ν_ℓ, the periods π/ν_2 and $2\pi/\nu_2$, and the coefficients of Appendix E) are not from runs.

TABLE II. Provenance. Reduced model at $\Gamma = \sqrt{18}$, $n_{sp} = 1$, seed on the asymptote, unless stated. Run keys: [P] `2026-10-18-pi-chatgpt-independent-verification`; [E] `2026-10-19-gpt-e-similariton`; [V] `2026-10-19-claude-interactive.13-similariton-lo-verification`; [A] `2026-10-11-gpt-a-highlights-audit-c2`; [Q] `2026-10-19-claude-interactive-gpt6pro-quick-repro`; [S] `2026-10-08-claude-a-amp-sim-checks`.

Quantity	Value	Run	Agent
Shape modes (Sec. III)			
Fitted powers of $N_{\mathrm{out}}\mathrm{Var}\, n_2$, $\varepsilon_0^2 N_{\mathrm{out}}\mathrm{Var}\, p_2$, $12 \le \tau \le 26$; between equal phases	0.32461, 1.69203; 0.33349, 1.66676	P	GPT 6 Pro
Compensated $\ell = 2$ variances, range over $12 \le \tau \le 30$	[1.11, 2.68]; [0.186, 0.447] (fig.)	E	gpt-e
Full NLSE $\ell = 2$ decay rate, log-gain frequency ($\varepsilon_0 = 0.05$, $1 \le \tau \le 6$); change on refinement	0.333825, 0.470401; 2×10^{-10}	P	GPT 6 Pro
Full-field rates $\ell = 2, 3, 4$ ($\varepsilon_0 = 0.05$, $gL = 6$)	0.33316, 0.33375, 0.33259	A	gpt-a
Spectrum and matched LO (Sec. IV)			
Fitted powers of μ_2/μ_1, $1 - r^2$ ($12 \le \tau \le 26$); $1 - r^2$ between equal phases	−0.66551, −0.65351; −0.66734	P	GPT 6 Pro
μ_2/μ_1, $1 - r^2$ at $\tau = 12$	3.03×10^{-3}, 8.96×10^{-5}	P, A, E	GPT 6 Pro, gpt-a, gpt-e
Fitted power of $\varepsilon_0^2 C_{PP}$, $12 \le \tau \le 26$	2.3335	P	GPT 6 Pro
Same, $12 \le \tau \le 26$ and 26–40; held out $\Gamma = 1$ (unchirped seed)	2.33350, 2.333333; 2.33175, 2.333314	V	claude-interactive.13
Error exponent of the C_{PP} prefactor, $\Gamma = \sqrt{18}$, $\Gamma = 1$; ratio at $\tau = 2, 4, 6$ ($\Gamma = \sqrt{18}$; $\Gamma = 1$)	−0.334, −0.332; 1.098, 1.106, 1.024; 0.766, 0.855, 0.935	V	claude-interactive.13
Error of Eq. (6): reduced; full field	7×10^{-15}; order 10^{-11}	V	claude-interactive.13
Full NLSE, complete noise, $\tau = 4$, $\varepsilon_0 = 0.05, 0.025$: μ_2/μ_1; $1 - r^2$; leading-order values	0.31249, 0.29547; 0.02920, 0.02841; about 0.278, 0.02804	P	GPT 6 Pro
Squeezing (Sec. V)			
Initial slope against $2n_{sp}g(1 - R)$; root against $R = 1$; commutator	7×10^{-7}; 3.3×10^{-9}; 4×10^{-13}	V	claude-interactive.13
Held-out $n_{sp} = 2$ thresholds $\gamma P_0/g$: Gaussian; parabola at $\delta = 0.05, 0.01, 0.005$	2.828427; 2.50699, 2.50030, 2.500076	V	claude-interactive.13
Counterexample λ_{LO}, $\tau = 0.02$, $\varepsilon_0 = 0.05$	0.8964864655	P, Q	GPT 6 Pro; claude-interactive
Same, verifier's code	0.8964864646–0.8964864653	V	claude-interactive.13
Complete-ASE matched-mode minimum, $\tau = 0.2$, $\varepsilon_0 = 0.05$ (clean-context audit)	0.436717	E	gpt-e
Length over which the mode stays squeezed ($\varepsilon_0 = 0.05$)	$\tau \approx 1.6$	V	claude-interactive.13
Second-order coefficients at $R = 1$: smoothed parabola $n_{sp} = 1, 2$; Gaussian $n_{sp} = 1, 2$	−0.85, −1.41; −0.69, −0.76	V	claude-interactive.13
Truncated Wigner, $N_0 = 10^6$, Gaussian, $\tau = 0.3$: λ_{LO} against linearized	0.4729 ± 0.0013 vs 0.4729	V	claude-interactive.13
Truncated Wigner at $N_0 = 10^4$; 10^3	0.488–0.492; 0.63–0.69	V	claude-interactive.13
$G^{-1/3}\lambda_{\min}$: $12 \le \tau \le 26$; 26–80; $\Gamma = 1$	[0.338, 0.633]; [0.338, 0.625]; [0.247, 0.797]	V	claude-interactive.13
Full field λ_{LO} at $\tau = 4$, $\varepsilon_0 = 0.05$; excess over reduced at $\varepsilon_0 = 0.05, 0.025$	3.1592357; 4.2%, 1.3%	V	claude-interactive.13
Limits and outlook (Secs. VI, VII)			
Φ_{NL}^2 at $\varepsilon_0 = 0.05$, $\tau = 12, 26$	4.44×10^8, 5.68×10^{16}	V	claude-interactive.13
rms phase at $N_0 = 10^4$, $\varepsilon_0 = 0.05$, $\tau = 12$	199 rad	P	GPT 6 Pro
Photons in a 500 pJ, 3 ps seed at 1.03 μm	2.6×10^9	S	claude-a
Held-out $\Gamma = 1$, $gL = 2$ spectral ratio at $\varepsilon_0 = 0.05, 0.025$ (table: 0.4257)	0.385841, 0.385432	A	gpt-a
μ_2/μ_1 at $\ln G = 12$ (outlook)	3.0×10^{-3}	P	GPT 6 Pro

[1] M. E. Fermann, V. I. Kruglov, B. C. Thomsen, J. M. Dudley, and J. D. Harvey, Self-similar propagation and amplification of parabolic pulses in optical fibers, Phys. Rev. Lett. **84**, 6010 (2000).

[2] V. I. Kruglov, A. C. Peacock, J. D. Harvey, and J. M. Dudley, Self-similar propagation of parabolic pulses in normal-dispersion fiber amplifiers, J. Opt. Soc. Am. B **19**, 461 (2002).

[3] C. Finot, G. Millot, and J. M. Dudley, Asymptotic characteristics of parabolic similariton pulses in optical fiber amplifiers, Opt. Lett. **29**, 2533 (2004).

[4] S. Wabnitz, Analytical dynamics of parabolic pulses in nonlinear optical fiber amplifiers, IEEE Photon. Technol. Lett. **19**, 507 (2007).

[5] S. Wabnitz, Analytical dynamics of optical similaritons, in *2007 Quantum Electronics and Laser Science Conference (QELS)* (2007) pp. 1–2.

[6] J. Zweck and C. R. Menyuk, Computation of the timing jitter, phase jitter, and linewidth of a similariton laser, J. Opt. Soc. Am. B **35**, 1200 (2018).

[7] S. Wang, B. Liu, Y. Song, and M. Hu, Intensity noise reduction of a high-power nonlinear femtosecond fiber amplifier based on spectral-breathing self-similar parabolic pulse evolution, in *Laser Sources and Applications III*, Proc. SPIE, Vol. 9893 (2016) p. 98930J.

[8] S. Wang, W. Chen, P. Qin, Y. Song, M. Hu, and B. Liu, Spectral and temporal breathing self-similar evolution in a fiber amplifier for low-noise transform-limited pulse generation, Opt. Lett. **41**, 5286 (2016).

[9] N. Haviv, M. Kozlov, M. Krüger, and P. Sidorenko, Noise suppression in gain-managed nonlinear amplifiers, Opt. Lett. **51**, 1812 (2026).

[10] C. Menotti and S. Stringari, Collective oscillations of a one-dimensional trapped Bose-Einstein gas, Phys. Rev. A **66**, 043610 (2002), cond-mat/0201158.

[11] P. O. Fedichev and U. R. Fischer, "cosmological" quasiparticle production in harmonically trapped superfluid gases, Phys. Rev. A **69**, 033602 (2004), cond-mat/0303063.

[12] C. M. Caves, Quantum limits on noise in linear amplifiers, Phys. Rev. D **26**, 1817 (1982).

[13] P. D. Drummond and J. F. Corney, Quantum noise in optical fibers. I. Stochastic equations, J. Opt. Soc. Am. B **18**, 139 (2001), quant-ph/9912095.

[14] J. P. Gordon and L. F. Mollenauer, Phase noise in photonic communications systems using linear amplifiers, Opt. Lett. **15**, 1351 (1990).

[15] K.-P. Ho and J. M. Kahn, Electronic compensation technique to mitigate nonlinear phase noise, J. Lightwave Technol. **22**, 779 (2004), physics/0211097.

[16] M. Shirasaki and H. A. Haus, Squeezing of pulses in a nonlinear interferometer, J. Opt. Soc. Am. B **7**, 30 (1990).

[17] C. R. Doerr, M. Shirasaki, and F. I. Khatri, Simulation of pulsed squeezing in optical fiber with chromatic dispersion, J. Opt. Soc. Am. B **11**, 143 (1994).

[18] Y.-L. Ren, R.-T. Cao, S.-L. Ma, R. Zhang, F.-L. Li, F. Nori, and P.-B. Li, Surpassing quantum noise limits with nonlinear amplification (2026), arXiv:2603.11574.